St.Philomena's College (Autonomus), Mysore

PG Department of Mathematics

Question Bank (Revised Curriculum 2018 onwards)

Second Year - Third Semester (2019 -21 Batch)

Course Title (Paper Title): Graph Theory Q.P.Code-57203

Unit	S.No	Question	Marks
1	1	Define order and size of a graph. Find order and size of complete graph.	2m
1	2	Draw all graphs with four points.	2m
1	3	Define multi-graph and pseudograph with example.	2m
1	4	What is the maximum number of lines in a graph with p points? Justify.	2m
1	5	Draw two different (5,5) graph.	2m
1	6	Draw a graph with point representing the numbers $1, 2, 3, ..., 10$ in which two points are adjacent if and only if they have a common divisor greater than one.	2m
1	7	Define degree of a point in a graph. Draw all graphs on $p = 3$ points.	2m
1	8	Define minimum degree of a graph with example.	2m
1	9	Define Maximum degree of a graph with example.	2m
1	10	Define regular graph. Draw all regular graph with four points.	2m
1	11	Define cubic graph. Draw all cubic graph with eight points.	2m

1	12	Define complete graph with example.	2m
1	13	Let G be a (p,q) graph all of whose points have degree k or k+1. If G has $t > 0$ points of degree k then show that t=p(k+1)-2q.	2m
1	14	Show that number of lines of an r-regular graph with p points is $q = \frac{pr}{2}$.	2m
1	15	Define isomorphism of two graphs. Draw all non-isomorphic graphs with three points.	2m
1	16	Define complement and self complement of a graph with example.	2m
1	17	Let G be a (p,q) graph then prove that for $v \in V(G)$ $deg_{\overline{G}}v = p - 1 - deg_G v$.	2m
1	18	Show that a (p,q) graph is a complete graph if and only if $q = \frac{p(p-1)}{2}$.	2m
1	19	Find the complement of K_p and show that K_p is a (p-1)regular graph.	2m
1	20	Define spanning subgraph of a graph. Draw all spanning subgraph of K_3.	2m
1	21	Define induced subgraph of a graph. Draw two induced subgraph of K_5.	2m
1	22	Is complement of a complete graph is a subgraph of that graph? Justify.	2m
1	23	Define complete bipartite graph. Give an example of a bipartite graph which is regular.	2m
1	24	Show that every disconnected graph has at least two components.	2m
1	25	Define distance between two points with example.	2m

1	26	Express the wheel graph and $K_4 - e$ graph, where $e \in E(K_4)$ as join of two graphs.	2m
1	27	Define degree sequence of a graph with example.	2m
1	28	Check whether the sequence (5,3,3,3,3,2,2,1,0) is graphical. If so construct the graph.	2m
1	29	Define union of two graphs with example.	2m
1	30	Define join of two graphs with example.	2m
1	31	Define product of two graphs with example.	2m
1	32	Define composition of two graphs with example.	2m
2	33	Define tree. Draw all trees with four points.	2m
2	34	Draw all trees with seven points and $\Delta(T) \geq 4$.	2m
2	35	For any connected graph G prove that $r(G) \leq diam(G) \leq 2r(G)$.	2m
2	36	Define cut point of a graph with example.	2m
2	37	Define bridge point of a graph with example.	2m
2	38	If v is a cut point of G then show that v is not a cut point of $\overline{G}$.	2m
2	39	Define block and end block of a graph with example.	2m
2	40	Define block index of point in a graph and find the block index of an isolated point and a cut point.	2m
2	41	Define block graph of a graph. Draw block graph of $K_{1,6}$	2m

2	42	Draw two different graphs G_1 and G_2 such that block graph of G_1 is isomorphic to block graph of G_2.	2m
2	43	Define cut point graph of a graph with example.	2m
3	44	Define point connectivity of a graph and find the point connectivity of $K_{m,n}$.	2m
3	45	Define line connectivity of a graph find the line connectivity of cycle with p number of points.	2m
3	46	Let G be a graph with $\lambda(G) = 1$ then what are the possible values of $\lambda(G - v)$ and $\lambda(G - e)$ where, $v \in V(G)$, $e \in E(G)$.	2m
3	47	Let G be a graph with $\kappa(G) = 1$ then what are the possible values of $\kappa(G - v)$ and $\kappa(G - e)$ where, $v \in V(G)$, $e \in E(G)$.	2m
3	48	Construct a graph with $\kappa(G) = 3$ $\lambda(G) = 4$ and $\delta(G) = 5$.	2m
3	49	Define Eulerian graph with example.	2m
3	50	Define hamiltonian. Give an example for a non-hamiltonian graph which contains a hamiltonian path.	2m
3	51	Determine whether the degree sequences G=(6,4,4,2,2,2,2,2,2)is Eulerian. If so find a partition of lines of G into cycles.	2m
3	52	Show that any hamiltonian graph has no cut points.	2m
3	53	Define neighborhood of a point with example.	2m
3	54	Define maximal non-hamiltonian graph with example.	2m

3	55	Define closure of a graph with example.	2m
3	56	If G is a graph with $p \geq 3$ points such that $deg(v) \geq \frac{p}{2}$ for every point in G then prove that G is hamiltonian.	2m
4	57	Define planar graph with example.	2m
4	58	Define maximal planar graph with example.	2m
4	59	Give an example for a for a graph G such that both G and $\overline{G}$ are planar.	2m
4	60	Define regular polyhedron with example.	2m
4	61	Define crossing number of a graph with example.	2m
4	62	Define outer planar graph with example.	2m
4	63	Draw all maximal outer planar graphs with seven points.	2m
4	64	State Euler's polyhedron formula.	2m
4	65	Define the point covering number of a graph with example.	2m
4	66	Define the line covering number of a graph with example.	2m
4	67	Define the point independence number of a graph with example.	2m
4	68	Define chromatic number of a graph with example.	2m
4	69	Find the chromatic number of complete bipartite and wheel graphs.	2m
4	70	If H is a subgraph of G then prove that $\chi(G) \geq \chi(H)$.	2m
1	71	State and prove First theorem of graph theory.	4m
1	72	Prove that every cubic graph has even number of points.	4m

1	73	Prove that any induced subgraph of a complete graph is complete.	4m
1	74	Find the number of points and lines in complete bipartite graph.	4m
1	75	Let G be a (p,q) graph. If G is bipartite then show that $q \leq \frac{p^2}{4}$.	4m
3	76	If G is hamiltonian graph then prove that for every non empty proper subset S of V(G), $w(G_S) \leq \|S\|$, where w(G-S) denotes the number of components in any graph G-S.	4m
3	77	Prove that a graph is Hamiltonian if and only if its closure is Hamiltonian.	4m
4	78	If G is a planar (p,q) graph with r-regions and k-components, then prove that p-q+r=k+1.	4m
4	79	If G is a maximal planar (p,q) graph with $p \geq 3$ then prove that q=3p-6.	4m
4	80	If G is a planar (p,q) graph with $p \geq 3$ then prove that $q \leq 3p - 6$.	4m
4	81	Prove that every planar graph contain a point of degree at most 5.	4m
4	82	Prove that the graph K_5 and $K_{3,3}$ are non-planar.	4m
4	83	If G is a planar (p,q) graph without triangles and $p \geq 3$ then prove that $q \leq 2p - 4$.	4m
4	84	Prove that at least one face of every polyhedron is bounded by n-cycle for some n=3,4,5,...	4m
4	85	If G is a(p,q) planar graph in which every region is a n-cycle, then prove that $q = \frac{n(p-2)}{n-2}$.	4m

4	86	If G is a (p,q) graph then prove that $\chi(G) \geq \frac{p^2}{p^2 - 2q}$.	4m
1	87	Explain Konigsberg bridge problem.	5m
1	88	Prove that in any graph the number of point of odd degree is even.	5m
1	89	Prove that every graph with at least two points contains two points of the same degree.	5m
1	90	Let G be a (p,q) graph then prove that $\delta \leq \frac{2q}{p} \leq \Delta$. Does there exist a 3- regular graph with six points? If so construct the graph.	5m
4	91	Prove that for any graph G, $\chi(G) \leq 1 + max\delta(G')$ where the maximum is taken over all induced subgraphs G' of G.	5m
4	92	For any graph G prove that $\frac{p}{\beta} \leq \chi(G) \leq p - \beta + 1$. where β is the independence number of G.	5m
1	93	Prove that if G is regular then $\overline{G}$ is also regular.	6m
1	94	Prove that for any graph with six points G or $\overline{G}$ contains a triangle.	6m
1	95	Prove that a non trivial graph is bipartite if and only if all its cycles are even.	6m
1	96	Prove that G or $\overline{G}$ is connected for any graph G.	6m
2	97	Prove that every non trivial tree has at least two points of degree one.	6m

2	98	Let v be a point of a connected graph G then prove that the following statements are equivalent i) v is a cut point of G. ii) There exist a partition of $V - \{v\}$ into two subsets U and W such that for any point $u \in U$ and $w \in W$ the point v lies on every u-w path. iii) There exist two points u and w distinct from v such that v is on every u-w path.	6m
2	99	Prove that every non-trivial connected graph with p points has at most p-2 cut points	6m
2	100	For any graph G prove that $b(G) = c(G) + \sum_{v \in V(G)} [b(v) - 1]$ where, b(G) is the number of blocks of G and c(G) is the number of components of G.	6m
2	101	Prove that a graph H is the block graph of some graph if and only if every block of H is complete.	6m
3	102	Prove that among all graphs with p points and q lines, the maximum connectivity is zero when $q < p - 1$ and is $\left\lfloor \frac{2q}{p} \right\rfloor$ when $q \geq p - 1$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x.	6m
3	103	Prove that if G is a k-connected (p,q) graph then $q \geq \frac{pk}{2}$. Check whether there is 5-connected graph with 20 lines.	6m
3	104	Prove that in a connected graph G there is a Eulerian trail if and only if the number of points of odd degree is either zero or two.	6m
4	105	State and prove Euler's formula of a connected planar graph.	6m

4	106	Prove that the graph K_5 and $K_{3,3}$ are non-planar.	6m
4	107	Let G be a maximal planar graph of order $p \geq 4$ and let p_i denote the number of points of degree i, for $i = 3, 4, ..., n = \Delta(G)$ then prove that $3p_3 + 2p_4 + p_5 = p_7 + 2p_8 + \cdots + (n-6)p_n + 12$	6m
4	108	Prove that the following graphs are equivalent for any graph G (i) G is 2-colorable. (ii) G is bipartite.	6m
4	109	Prove that there exists a k-coloring of a graph G if and only if V(G) can be partitioned into k subsets $V_1, V_2, ..., Vk$ such that no two points in V_i, i=1,2,...,k are adjacent.	6m
1	110	Prove that any self complementary graph has 4n or 4n+1 points.	7m
1	111	If a graph has exactly two points of odd degree then prove that there exists a path joining these two points.	7m
1	112	Prove that every u-v walk in a graph G contains a u-v path.	7m
1	113	Prove that a closed walk of odd length contains a cycle.	7m
1	114	Prove that a graph with p points and $\delta \geq \frac{p-1}{2}$ is connected graph.	7m
1	115	Prove that a graph G is disconnected if and only if its point set V can be partitioned into two non empty disjoint subsets V_1 and V_2 such that there exist no line e=uv where $u \in V_1$ and $v \in V_2$.	7m
1	116	Prove that if a (p,q) graph is connected then $q \geq p - 1$.	7m

1	117	Prove that a graph with p points and k-components has at most $\frac{(p-k)(p-k+1)}{2}$ lines.	7m
2	118	Prove that every tree has a center consisting of either one point or two adjacent points.	7m
2	119	Prove that every connected graph has a spanning tree.	7m
3	120	Let G be a connected graph with 2n points, $n \geq 1$ odd points, then prove that the set of lines of G can be partitioned into n open trials.	7m
3	121	If every block of a connected graph G is Eulerian, then G is Eulerian.	7m
1	122	Prove that a graph $G = (V, E)$ is connected if and only if for any partition of V into subsets V_1 and V_2 there is a line of G joining a point of V_1 to a point of V_2.	8m
1	123	Let G_1 be a (p_1, q_1) graph and G_2 be a (p_2, q_2) graph then prove that i) $G_1 \cup G_2$ is a $(p_1 + p_2, q_1 + q_2)$ graph. ii) $G_1 + G_2$ is a $(p_1 + p_2, q_1 + q_2 + p_1p_2)$ graph. iii) $G_1 \times G_2$ is a $(p_1p_2, q_1p_2 + q_2p_1)$ graph. iv) $G_1[G_2]$ is a $(p_1p_2, p_1q_2 + p_2^2q_1)$ graph.	8m
1	124	State and prove Havel-Hakimi theorem.	8m
2	125	Let G be a (p,q) graph then prove that the following statements are equivalent i) G is a tree. ii) Every two points of G are joined by unique path. iii) G is connected and p=q+1. iv) G is acyclic and p=q+1.	8m

2	126	Prove that a line x of a connected graph G is a bridge if and only if it is not on any cycle of G.	8m
2	127	Prove that a cubic graph has a cut point if and only if it has a bridge.	8m
2	128	Let G be a connected graph with at least three points then prove that the following statements are equivalent i) G is a block. ii) Any two points of G lie on a common cycle. iii) Any point and any line of G lie on a common cycle. iv) Any two lines of G lie on a common cycle.	8m
2	129	Let G be a connected graph with at least three points then prove that the following statements are equivalent i) Every two lines of G lie on a common cycle. ii) Given two points and one line of G there is a path joining the points which contains the line. iii) For every three distinct point of G there is a path joining any two of them which contains the third. iv) For every three distinct point of G there is a path joining any two of them which doesn't contains the third.	8m
3	130	For any graph G prove that $\kappa(G) \leq \lambda(G) \leq \delta(G)$.	8m
3	131	If graph G has p points and $\delta(G) \geq \frac{p}{2}$ then prove that $\lambda(G) = \delta(G)$.	8m
3	132	Prove that the following statements are equivalent for a connected graph G, (i) G is Eulerian. (ii) Every point of G has even degree. (iii) The set of lines of G can be partitioned into cycles.	8m

4	133	For any graph G, the sum and product of χ and $\overline{\chi}$ satisfy the inequalities $2\sqrt{p} \leq \chi + \overline{\chi} \leq p+1$ and $p \leq \chi\overline{\chi} \leq \frac{(p+1)^2}{4}$.	8m
4	134	State and prove five color theorem.	8m
1	135	Prove that the maximum number of lines among all p point graphs with no triangles is $\left\lfloor \frac{p^2}{4} \right\rfloor$ where, $\lfloor x \rfloor$ denotes the greatest integer not exceeding x.	10m
3	136	If G is a graph with $p \geq 3$ points and $\delta(G) \geq \frac{p}{2}$ then prove that G is hamiltonian.	10m
3	137	Let G be a graph with p points and let u and v be non-adjacent points in G such that $deg(u) + deg(v) \geq p$. Then prove that G is hamiltonian if and only if G+uv is hamiltonian.	10m
4	138	Prove that there are five regular polyhedron.	10m
2	139	State and prove Menger's theorem.	14m

Model Question Paper

Q.P Code: 57203

St. Philomena's College (Autonomous) Mysore

III Semester M.Sc. Final Examination : December - 2019

Subject: MATHEMATICS

Title: Graph Theory (SC)

Time: 3 Hours **Max Marks: 70**

PART - A

Answer the following questions: **7×2=14**

1. a. Define tree. Draw all trees with 5 points.
   b. Define Eccentricity of a vertex. Find the radius of K_p.
   c. Define spanning tree. Find all distinct spanning trees of $K_{2,2}$.
   d. Define an Eulerian graph. Give an example.
   e. Define vertex degree. Draw all graphs on $p = 3$ points.
   f. Prove that if G is a K – connected (p, q) graph then $q \geq \frac{pk}{2}$.
   g. Define chromatic number of a graph. Find the chromatic number of C_n.

PART –B

Answer the following:

2. a. State and prove Fundamental theorem of graph theory. And prove that in every graph the number of vertices of odd degree is even. **10**
   b. Define union and join of two graphs. Give examples. **04**

OR

3. a. Prove that a graph is bipartite if and only if all its cycles are even. **08**
   b. Prove that for any self complementary graph has $4n$ or $4n + 1$ points. **06**
4. a. Prove that a (p,q) graph is a tree if and only if it is connected and $p = q + 1$. **08**
   b. Prove that non-trivial tree has atleast two points of degree 1. **06**

OR

PTO

5. a. Prove that cubic graph has a cut point if and only if it has a bridge. **06**

b. Let v be a point of a connected graph G. Then the following statements are equivalent: **08**

i) v is a cutpoint of G

ii) there exist a partition of $V-\{v\}$ into two subsets V and W such that for any point $u \in U$ and $w \in W$, the point v lies on every $u-w$ path.

iii) There exist two point u and w distinct from v such that v is on every $u-w$ path.

6. a. Prove that for any graph G, $K(G) \leq \lambda(G) \leq \delta(G)$. **08**

b. Prove that there is no 3 – connected graph with 7 lines. **06**

OR

7. a. Prove that if G is a graph with $p \geq 3$ point and $\delta(G) \geq p/2$ then G is Hamiltonian. **08**

b. Find the chromatic number of the following graphs: **06**

i) Cp ii) $\overline{K_{m,n}}$ iii) K_p

8. a. Prove that if G is a connected plane graph with p vertices q edges and r – regions then $p-q+r=2$. **08**

b. Show that the graph K_5 and $K_{3,3}$ are non-planar. **06**

OR

9. a. Prove that for any graph $x(G)=1+\max \delta(G^1)$ where the maximum is taken over all induced sub-graphs G^1 of G. **08**

b. Prove that if G is (p, q) graph then $X(G) \geq \dfrac{P^2}{P^2-2q}$. **06**

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